

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Quadratic Equations

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Solve $x^2 - 7x + 12 = 0$.

Answer:

Factorising gives $(x - 3)(x - 4) = 0$. Hence $x = 3$ or $x = 4$.

Question 2

Solve $x^2 - 9x + 14 = 0$.

Answer:

Factorising gives $(x - 2)(x - 7) = 0$. Hence $x = 2$ or $x = 7$.

Question 3

Solve $x^2 - 2x - 15 = 0$.

Answer:

Factorising gives $(x - (-3))(x - 5) = 0$. Hence $x = -3$ or $x = 5$.

Question 4

Solve $x^2 - 4x - 12 = 0$.

Answer:

Factorising gives $(x - 6)(x - (-2)) = 0$. Hence $x = 6$ or $x = -2$.

Question 5

Solve $x^2 - 9x + 8 = 0$.

Answer:

Factorising gives $(x - 8)(x - 1) = 0$. Hence $x = 8$ or $x = 1$.

Question 6

Solve $x^2 + 9x + 20 = 0$.

Answer:

Factorising gives $(x - (-4))(x - (-5)) = 0$. Hence $x = -4$ or $x = -5$.

Question 7

Solve $x^2 - 8x - 9 = 0$.

Answer:

Factorising gives $(x - 9)(x - (-1)) = 0$. Hence $x = 9$ or $x = -1$.

Question 8

Solve $x^2 - 12x + 20 = 0$.

Answer:

Factorising gives $(x - 10)(x - 2) = 0$. Hence $x = 10$ or $x = 2$.

Question 9

Solve $x^2 - 10x + 25 = 0$.

Answer:

Factorising gives $(x - 5)(x - 5) = 0$. Hence $x = 5$ or $x = 5$.

Quadratic Equations - Continued

Question 10

Solve $x^2 - 4x - 21 = 0$.

Answer:

Factorising gives $(x - 7)(x - (-3)) = 0$. Hence $x = 7$ or $x = -3$.

Question 11

Find the nature of roots of $1x^2 + 2x + 5 = 0$.

Answer:

$D = b^2 - 4ac = -16$. Therefore the equation has no real roots.

Question 12

Find the nature of roots of $2x^2 - 4x + 5 = 0$.

Answer:

$D = b^2 - 4ac = -24$. Therefore the equation has no real roots.

Question 13

Find the nature of roots of $1x^2 - 6x + 9 = 0$.

Answer:

$D = b^2 - 4ac = 0$. Therefore the equation has equal real roots.

Question 14

Find the nature of roots of $3x^2 + 4x + 1 = 0$.

Answer:

$D = b^2 - 4ac = 4$. Therefore the equation has two distinct real roots.

Question 15

Find the nature of roots of $2x^2 + 5x + 2 = 0$.

Answer:

$D = b^2 - 4ac = 9$. Therefore the equation has two distinct real roots.

Question 16

A rectangle has area 48 m^2 and its length is 2 m more than breadth. Find dimensions.

Answer:

Let breadth = x and length = $x + 2$. From $x(x+2) = 48$, $x = 6$. Thus breadth = 6 m and length = 8 m.

Question 17

A rectangle has area 60 m^2 and its length is 4 m more than breadth. Find dimensions.

Answer:

Let breadth = x and length = $x + 4$. From $x(x+4) = 60$, $x = 6$. Thus breadth = 6 m and length = 10 m.

Question 18

A rectangle has area 96 m^2 and its length is 4 m more than breadth. Find dimensions.

Answer:

Let breadth = x and length = $x + 4$. From $x(x+4) = 96$, $x = 8$. Thus breadth = 8 m and length = 12 m.

Question 19

A rectangle has area 80 m^2 and its length is 2 m more than breadth. Find dimensions.

Answer:

Let breadth = x and length = $x + 2$. From $x(x+2) = 80$, $x = 8$. Thus breadth = 8 m and length = 10 m.

Quadratic Equations - Continued

Question 20

A rectangle has area 63 m^2 and its length is 2 m more than breadth. Find dimensions.

Answer:

Let breadth = x and length = $x + 2$. From $x(x+2) = 63$, $x = 7$. Thus breadth = 7 m and length = 9 m.

Question 21

State the main formula, rule or theorem used in Quadratic Equations.

Answer:

For $ax^2+bx+c=0$, $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$.

Question 22

What do the important symbols or quantities in Quadratic Equations represent?

Answer:

The discriminant $D=b^2-4ac$ determines the nature of roots.

Question 23

Write the correct method for solving a Quadratic Equations numerical.

Answer:

Bring the equation to standard form, factorise or use the formula, then reject impossible values in word problems.

Question 24

Which unit, sign convention or condition must be checked in a Quadratic Equations answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Quadratic Equations be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Quadratic Equations questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Quadratic Equations numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Quadratic Equations?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Quadratic Equations - Continued

Question 29

How can a student decide which formula to use in Quadratic Equations?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Quadratic Equations?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY